Dependent paths in a family of function types are equivalent to pointwise identifications

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled thm:dpath-arrow (source lines 1660-1671); the proof span 1672-1675 is the author's proof environment, retained line for line.

Editorial note. The author proof reduces the equivalence to function extensionality by path induction and derives the moreover clause from the computation rule for function extensionality; no step is deferred to the reader.

Editorial note. The second decomposition item also cites the author's transport-arrow equation, retained as context, and the two lemmas declared as external refs.

Editorial note. Revision history: r001 assembly was refused because the external refs were flagged load-bearing and r002 because their kinds were outside the policy vocabulary; r003 restates the refs as reviewed navigation pointers with policy-vocabulary kinds, without changing the retained spans.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the characterization of dependent paths in families of function types. The counted lemma and its author proof are retained; the introductory transport-arrow equation is retained as context.

Allowed standard prerequisites (editorial).

- Type families A and B over a type X, functions between their fibers, and the identity type of X with its reflexivity element.
- Transport along a path in a type family, applied to functions and to elements.
- Path induction with its conversion rule.
- Function extensionality: the equivalence between pointwise equalities of functions and identifications of functions, together with its computation rule.

Source reference. Original lemma 2.3.9: sources/hott-basics/pinned-original-source.tex, source lines 951–951. Author lemma asserted at source lines 951-954 (book 2.3.9), cited by the second decomposition item of the retained statement. Navigation pointer only: the counted statement and proof are retained verbatim and no part of them is replaced by the reference; the author uses the cited lemma as established.

Source reference. Original lemma 2.1.4: sources/hott-basics/pinned-original-source.tex, source lines 393–393. Author lemma proved in place in the source (book 2.1.4) and retained in-corpus as a sibling card; cited by the second decomposition item of the retained statement. Navigation pointer only: the counted statement and proof are retained verbatim and no part of them is replaced by the reference.

Given a type X, a path $p : x_1 =_X x_2$, type families $A, B : X \to \mathcal{U}$, and a function $f : A(x_1) \to B(x_1)$, we have

$$\mathsf{transport}^{A\to B}(p, f) = \Big(x \mapsto \mathsf{transport}^{B}(p, f(\mathsf{transport}^{A}(p^{-1}, x)))\Big) \tag{0.0.1}$$

where $A \to B$ denotes abusively the type family $X \to \mathcal{U}$ defined by

$$(A \to B)(x) :\equiv (A(x) \to B(x)).$$

Lemma 0.0.2. *Given type families $A, B : X \to \mathcal{U}$ and $p : x =_X y$, and also $f : A(x) \to B(x)$ and $g : A(y) \to B(y)$, we have an equivalence*

$$\big(p_*(f) = g\big) \simeq \prod_{a:A(x)} (p_*(f(a)) = g(p_*(a))).$$

Moreover, if $q : p_*(f) = g$ corresponds under this equivalence to $\widehat{q}$, then for $a : A(x)$, the path

$$\mathsf{happly}(q, p_*(a)) : (p_*(f))(p_*(a)) = g(p_*(a))$$

is equal to the concatenated path $i \centerdot j \centerdot k$, where

- *$i : (p_*(f))(p_*(a)) = p_*\big(f({p^{-1}}_*(p_*(a)))\big)$ comes from* (0.0.1)*,*
- *$j : p_*\big(f({p^{-1}}_*(p_*(a)))\big) = p_*(f(a))$ comes from original lemmas 2.3.9 and 2.1.4, and*
- *$k : p_*(f(a)) = g(p_*(a))$ is $\widehat{q}(a)$.*

Proof. By path induction, we may assume p is reflexivity, in which case the desired equivalence reduces to function extensionality. The second statement then follows by the computation rule for function extensionality. □